

Shwetanshu Sood

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EDUCATION

- Indian Institute of Technology, Mandi** Mandi, India
MBA (Data Science & AI); GPA: 8.7/10 2025 - Present
- Chandigarh University** Chandigarh, India
B.E. (CSE with AIML); GPA: 8.59/10 2020 - 2024

SKILLS SUMMARY

- Data Science & AIML:** Python, Data Processing, Data Cleaning, Data Warehousing, ML Algorithms, Deep Learning, NLP, LLMs (OpenAI, Copilot, Claude, Llama)
- Frontend:** HTML, CSS, JavaScript, ReactJS
- Backend:** Flask, NodeJS, ExpressJS
- Databases:** MongoDB, SQL Server, MySQL, Oracle Server
- Cloud:** Microsoft Azure (Synapse, VM Instances, Server Setup, Blob Storage)
- Miscellaneous:** TensorFlow, PyTorch, Git, RestAPI

EXPERIENCE

- Tata Consultancy Services** Noida, Uttar Pradesh
Software Engineer (Full-time) Jun 2024 - Jul 2025
 - Data Pipeline Automation:** Designed and automated data analysis pipelines using Python (python-docx, openpyxl) for generating statistical reports and efficacy reports (Excel), improving reporting efficiency by ~40%.
 - Large-scale Data Analysis:** Extracted, analyzed, and validated large-scale datasets from SQL Server and Azure Synapse DB, enabling business teams to make data-driven decisions with 95% accuracy.
 - Template-agnostic Scripting:** Developed dynamic Python scripts capable of handling diverse report formats, reducing manual intervention by ~60%.
 - TCS Gems Award:** Recognized for commendable performance and delivering value-added solutions under tight deadlines.

PROJECTS

- Mini RAG Application (Retrieval-Augmented Generation, FastAPI, Qdrant, Gemini, Cohere):** Designed and deployed a full-stack Retrieval-Augmented Generation (RAG) system enabling document upload, semantic search, and citation-grounded LLM responses. Implemented chunking (1000 tokens, 150 overlap), 384-dimension embeddings using sentence-transformers, and cosine similarity search with Qdrant Cloud. Integrated Cohere Rerank to improve contextual relevance and Google Gemini for answer generation, achieving 80% accuracy on evaluation benchmark. Deployed backend on HuggingFace Spaces and frontend on Vercel.
- MoodWave Music Matcher (Deep Learning, Facial Emotion Recognition, Recommendation Systems):** Developed an AI-powered music recommendation platform integrating TensorFlow-based facial emotion recognition with personalized playlist mapping, improving recommendation accuracy by 15% and boosting user engagement by 35%. Designed secure REST APIs using Node.js/Express with JWT authentication, managed MongoDB schema for users and playlists, and contributed to UI evolution through collaborative development.
- Face Generator WebApp using GANs (Generative Adversarial Networks, Flask, Web Development):** Constructed and trained a GAN on the 200,000+ image CelebA dataset, achieving generation of high-resolution, realistic human faces with 95.6% accuracy. Engineered a user-friendly Flask web app for on-demand face generation, optimizing real-time performance.

PUBLICATIONS

- MoodWave Music Matcher: An Exploration of AI Driven Music Curation Based on Facial Analysis:** Research on integrating Face Emotion Detection with Music website to recommend songs based on mood. Published on IEEE Explore.
- Smart Security in Conjunction with IoT and Deep Learning:** Research on integrating IoT and Deep Learning for intelligent security systems. Published on ResearchGate.

LEADERSHIP & VOLUNTEER EXPERIENCE

- Coordinator, Industry Interaction Cell, IIT Mandi** Mandi, India
Responsible for conducting industry talks with Business Leaders. 2025 - Present
- IEEE Representative, Chandigarh University (Delhi Section)** Chandigarh, India
Elected for leadership and technical prowess; led a team of 5 during B.E. project. 2020 - 2024
- Training & Pre-Placements Class Representative** Chandigarh, India
Fostered peer development and career readiness; worked with top management on client decisions. 2024 - 2025